Practical Session: Differential Gene Expression Analysis (Part 2)

Course Instructor: Chioma Onyido

Differential Gene expression Analysis

- **Differential Gene Expression** (DGE) analysis aims to identify genes whose expression levels **vary significantly** between **two or more biological conditions** (e.g., tumor vs. normal).

Differential Gene expression Analysis

- For example, DGE helps researchers understand which genes are **upregulated** or **downregulated** in tumors compared to normal tissues.

- These genes can serve as **biomarkers**, **therapeutic targets** or help infer affected pathways.

RNA sequencing

- **RNA sequencing** (RNA-Seq) is a high-throughput method to measure **transcript abundance** across the entire genome.

- **Why?:** Unbiased approach, wide range, quantitative and versatile.

RECAP:

Key Concepts?

Practical Session:

DGE analysis using R

(check the website for the recording)

R Packages for DGE Analysis

- **DESeq2**
- **edgeR**
- **Limma**
- TCGAbiolinks
- biomaRt
- org.Hs.e.g.db

Framing biological questions:

Which genes are differentially expressed between …

Further Analysis

- **Functional Enrichment Analysis Tools**

- DAVID (https://david.ncifcrf.gov/)
- GSEA (Gene Set Enrichment Analysis) (https://www.gsea-msigdb.org/gsea/index.jsp)
- Enrichr (https://maayanlab.cloud/Enrichr/)
- ShinyGO

Further Analysis

- **Literature**

- **Compare with public datasets/ findings**: See if the same DEGs appear in related studies.

Further Analysis

➢ **Machine learning**

- Build models to test if DEGs can distinguish between conditions (e.g., case vs. control).

Further Analysis

- **Experimental Validation**

- **qPCR**: to check expression levels of key DEGs in independent samples.
- **Western blot / Immunohistochemistry (IHC):** Validate protein expression.
- **Functional validation:** Knockdown or overexpress key genes in cell culture to see their effect.

Further Reading/ References

- **Bioconductor (2025).** Analyzing RNA-seq data with DESeq2 https://bioconductor.org/packages/devel/bioc/vignettes/DESeq2/inst/doc/DESeq2.html
- **Ann and Robert.** How do Genes work? https://www.luriechildrens.org/en/specialties-conditions/pediatric-genetic-disorders/about-genes/
- **BYJU's**. Gene Regulation. https://byjus.com/biology/gene-regulation/
- **Nature Education**. Gene Expression. https://www.nature.com/scitable/topicpage/gene-expression-14121669/